



ALS (Amyotrophic Lateral Sclerosis)

ALS is a disease that occurs as a result of the gradual deterioration and death of nerve cells called motor neurons in the brain and spinal cord. These cells give movement commands to the muscles. When they become impaired, the muscles become weak and atrophy, and voluntary movements (such as walking, speaking, swallowing, and breathing) become difficult. The disease is progressive. Sensory functions, bowel and bladder control, cardiac muscle, and generally cognitive functions are not affected, while the eye muscles may be affected only at a very late stage or may remain almost entirely unaffected.

Approximately 2-6 new cases per 100,000 people are reported worldwide each year. In Europe, the prevalence is around 10-12 per 100,000. It most commonly begins between the ages of 45 and 75, with an average age of onset of approximately 55-65 years. It is slightly more common in men than in women; however, the ratio becomes equal in advanced age. Approximately 90-95% of cases are sporadic (random), while 5-10% are familial/genetic. The lifetime risk is approximately 1 in 350 for men and 1 in 400 for women.

The initial symptoms are often mild and unilateral: weakness of the thumb and muscle wasting in the hand, foot drop in the leg (tripping while walking), or difficulty speaking or swallowing. Muscle fasciculations, cramps, stiffness, and fatigue are also common. Patients may mistake these symptoms for being “overworked” or having a “pinched nerve.” The symptoms gradually spread to other parts of the body. In approximately one-third of patients, the disease begins in the speech and swallowing muscles; this form may progress more rapidly. Mild cognitive or behavioral changes may occur in approximately half of patients, and frontotemporal dementia accompanies the disease in 10–15% of cases. In the advanced stages, walking, speaking, swallowing, and breathing become impaired; the patient may become bedridden and may require respiratory support.

The exact cause is unknown. Genetic factors play a role in approximately 10% of cases; the most common genetic abnormalities include repeat expansion in the C9orf72 gene and mutations in SOD1, TARDBP, and FUS. In sporadic cases, age, male sex, smoking, high levels of physical activity, head trauma, chemical substances, heavy metals, and certain infections are considered potential risk factors, but a definitive association has not been established.

Upper and lower motor neurons in the motor cortex, brainstem, and anterior horns of the spinal cord are lost. In more than 95% of patients, a protein called TDP-43 accumulates within cells. Multiple pathways, including protein dysfunction, abnormalities in RNA metabolism, cytoskeletal dysfunction, and impaired axonal transport, play a role in the disease.

There is no single definitive test. The diagnosis is made through a neurological examination, EMG (evaluation of the electrical activity of nerves and muscles), and investigations such as blood tests, MRI, and lumbar puncture to exclude conditions that mimic the disease (nerve compression, disc herniation, muscle diseases, and certain cancers). The disease may be difficult to distinguish from other conditions in its early stages; therefore, careful evaluation is important.

There is no definitive cure. Riluzole may extend survival by an average of 3–6 months. In some countries, edaravone is used, and tofersen is used in patients with SOD1 mutations. The fundamental approach is multidisciplinary care: physical therapy, speech and swallowing therapy, respiratory support (non-invasive ventilation), and medications for symptom management. Good care, regular follow-up, and effective patient–physician communication can significantly affect both quality of life and survival.

After diagnosis, the average survival time is 2-5 years; 5-10% of patients live for 10 years or longer. Death most often occurs due to respiratory failure. Immobility, inadequate nutrition, and respiratory problems increase the risk of additional complications. Good care and supportive treatment can prolong this period.

Antisense oligonucleotide therapies targeting specific genetic subtypes and stem cell studies are ongoing. Research



into biomarkers and personalized approaches provides hope. The importance of multidisciplinary care is increasingly emphasized.

In summary, ALS is a disease characterized by the progressive loss of motor neurons in the brain and spinal cord, leading to muscle weakness and atrophy, and for which there is currently no definitive cure. It generally occurs in the 50s and is slightly more common in men; symptoms often begin with mild weakness in the hand or leg and gradually spread throughout the body. Its cause is largely unknown, although genetic factors play a role in approximately 10% of cases. The diagnosis is made based on clinical findings and EMG, and treatment is primarily based on supportive care. The average survival time is 2–5 years, although good care may prolong this period.

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